The Frohmanian Tethered Metriplectic World Model in Lean 4: Canonical Bracket Splitting, Encoding Obstructions, and the Newton Potential

Companion paper to `BenFrohman/metriplectic-world-model`

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Draft — tracks build in progress. This repository formalizes a *model*: the tethered metriplectic system, which adds a symmetric tether bracket and cubic damping of peak vorticity to 3D vorticity dynamics. It proves nothing about unmodified Navier–Stokes ($\kappa = 0$) and does not resolve the Clay Millennium problem.

Abstract

We report a Lean 4/Mathlib formalization (0 `sorry`; axioms only `propext`, `Classical.choice`, `Quot.sound`) of the author's tethered metriplectic model. Proved unconditionally: the unique antisymmetric/symmetric splitting of a bilinear bracket (`split_exists_unique`); a non-circular canonicity theorem forcing the metric bracket to the form $c\langle \Pi_u x, \Pi_u y\rangle$ (`canonical_metric_bracket`); a uniform vorticity ceiling for the comparison inequality $M' \le CM^2 - \kappa M^3$; a log-loss replacement for the stretching hypothesis giving a ceiling when $\kappa > 4C_{\mathrm{abs}}$; several encoding counterexamples; the Newton-potential identity $\Delta(N \star g) = -4\pi g$ on $\mathbb{R}^3$ for $g \in C_c^2$; and $\operatorname{curl} \mathrm{BiotSavart}\,\omega = \omega$ for smooth, compactly supported, divergence-free ω (`curlBiotSavartTest`). Conditional and open items are listed by their exact Prop names.

1 Naming

`Pi_u` is the author's own name (manuscript of 31 May 2026): the projection $\Pi_u x = x - (\langle u, x\rangle / \|u\|^2)u$ off $\mathrm{span}\{u\}$. All other identifiers defined in this repository are *coined in this build* (marked [coined]), including `canonical_metric_bracket`, kept by the author's choice and flagged as coined. Modules under `FrohmanianSymplecticTether/` and `TetherMayForm/` are ported verbatim from earlier repositories.

2 Manuscript framing

The May/June 2026 manuscript introduces a tethered Lyapunov functional S_ε and reduces regularity to the comparison ODE $y' = Cy^2 - \kappa'' y^3$ for the peak vorticity, through transport cancellation, viscous dissipation at the maximum, a Hölder–Sobolev gate, and Young absorption at $\varepsilon = \kappa/4$ (Proposition 2). Within the model, the cubic term comes from the tether. The manuscript's hypothesis (H5), a non-circular majorant for the vorticity, is the genuine gap; for $\kappa = 0$ it is the Clay

core. Here κ has units of time and breaks Navier–Stokes scaling, which is why the model is distinct from NS.

3 Brackets and canonicity (`Bracket.lean`, `Canonical.lean`)

Theorem 1. *`split_exists_unique`[coined]: every bilinear bracket is uniquely an antisymmetric part plus a symmetric part.*

Theorem 2. *`canonical_metric_bracket`[coined]: on a real inner product space, a bilinear form B that is symmetric (C1), degenerate along u (C2: $B(u,\cdot)=0$) and invariant under every linear isometry fixing u equals $c\,\langle \Pi_u x, \Pi_u y\rangle$. The hypotheses do not mention Π_u. `canonical_metric_bracket_sign`[coined]: if $B(x,x)\le 0$ then $c\le 0$. `Pi_u_form_symm_degen`[coined]: the forced form satisfies C1, C2.*

Scope: pointwise at one state; does not force the $|\omega|^2$ weight, the value of κ, or C3. Also proved: `energy_conserved`, `energy_const_along`, `entropy_production`, `gen_entropy_eq`, `entropy_monotone_along`[coi

4 Ceiling and stretching (`Ceiling.lean`, `AuditFixes.lean`)

Theorem 3. *`uniform_ceiling`[coined]: if $M' \le CM^2 - \kappa M^3$, $\kappa > 0$, then $M(t) \le \max(M(0), \max(C/\kappa, 0))$ for $t \ge 0$. Also `absorbing`, `ceiling_antitone_kappa`, `kappa_zero_no_ceiling`[coined] (at $\kappa = 0$, $1/(1-t)$ satisfies $M' \le M^2$ and blows up).*

Correction. The hypothesis `StretchingCZ` (a pure y^3 bound on stretching at the maximum) is *false as a sup-norm bound* for Navier–Stokes: the known bound carries a logarithmic loss. Results using it are relabeled as model-only. Its honest replacement is `StretchingLogLoss` $d\,C_1\,C_2$[coined]: $St \le C_1 y^3 + C_2 y^3 \log(1+y)$.

Theorem 4. *`maxpoint_rate_logLoss`[coined]: under `EnergyIdentity`, `StretchingLogLoss`, `ViscousAtMax`, `TetherFeedback`, `CrossControlled`,*

$$y' \le \left(C_1 + \tfrac{\kappa}{4} + \tfrac{4C_2^2}{\kappa}\right) y^2 - \left(\tfrac{\kappa}{4} - C_{\text{abs}}\right) y^3,$$

so with `uniform_ceiling` a uniform ceiling follows whenever $\kappa > 4C_{\text{abs}}$ (Young absorption at $\varepsilon = \kappa/4$).

5 Encoding obstructions (`ChoiceFree`, `HelmholtzObstruction`, `TestClass`)

- `not_streamFunctionRepModGrad`[coined]: `StreamFunctionRepModGrad` fails for every ω — representatives are determined only a.e.; uniqueness must be stated almost everywhere (open).
- `arnoldBracketOf_antisymm`, `entropyCasimir_arnold`[coined]: a choice-free Arnold bracket and a Casimir criterion.
- `not_curl_biotSavart_eq`[coined]: curl(BiotSavart ω) $= \omega$ fails for the point spike in the current encoding; `helicity_casimir_of_curlBiotSavart`[coined] is conditional on `CurlBiotSavartOn`.
- `IsTestOrbit`[coined] (smooth, compactly supported, div-free); `not_isTestOrbit_pointSpike`, `helicity_casimir_test`[coined] (conditional on `CurlBiotSavartTest`, discharged in `helicity_casimir_tes` Section 7); `integrableOn_ball_norm_rpow_neg`[coined].

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6 Newton potential on $\mathbb{R}^3$ (`NewtonKernel`, `RegularizedKernel`, `NewtonLaplacian`

Proved: `scalarLaplacian_newtonRadial` ($\Delta(1/|y|) = 0$, $y \neq 0$), `contDiff_newtonPotential`, `hasFDerivAt_newtonPotential` (differentiation under the integral); for $N_\varepsilon = (|y|^2 + \varepsilon^2)^{-1/2}$: `scalarLaplacian_regNewton` ($\Delta N_\varepsilon = -3\varepsilon^2(|y|^2+\varepsilon^2)^{-5/2}$), `regNewton_le`, `tendsto_regNewton`, `integral_neg_scalarLaplacian_regNewton` ($\int -\Delta N_\varepsilon = 4\pi$), `tendsto_regPeak_approxIdentity` (all [coined]). The regularized route avoids spherical Green identities, which Mathlib lacks. Step 3 is closed (`NewtonLaplacian`): `integral_mul_scalarLaplacian_eq` ($\int K\,\Delta g = \int \Delta K\, g$ for $K \in C^2$, $g \in C^2_c$, by Mathlib integration by parts, no boundary terms), `regNewton_conv_laplacian` ($N_\varepsilon \star \Delta g = (\Delta N_\varepsilon)\star g$), `tendsto_regNewton_conv` (the $\varepsilon \to 0$ limit, dominated by $N|h|$), `scalarLaplacian_newtonPoter` ($\Delta(N \star g) = N \star \Delta g$), and

Theorem 5. *`scalarLaplacian_newtonPotential_eq`[coined]: for $g \in C^2_c(\mathbb{R}^3)$ and $N(y) = 1/|y|$, $\Delta(N \star g)(x) = -4\pi g(x)$ for every x.*

7 Biot–Savart on the test class (`BiotSavartNewton`)

With $G_j(t) = t_j/|t|^3$ (`gradKer`[coined] at $\varepsilon = 0$):

Theorem 6. *`newtonPotential_deriv_eq`[coined]: for $h \in C^1_c$, $N \star \partial_j h = -\int G_j(t)\, h(x-t)\, dt$. (Regularized identity `regNewton_conv_deriv`, then $\varepsilon \to 0$ on both sides; the right side is dominated by $|t|^{-2}|h|$.)*

Theorem 7 (Step 5)**.** *`biotSavart_eq_curl_newtonVec`[coined]: for `IsTestOrbit` ω,* $\mathrm{BiotSavart}\,\omega = (4\pi)^{-1}\,\mathrm{curl}(N \star \omega)$ *pointwise, $N \star \omega$ taken componentwise (`newtonVec`).*

Theorem 8 (Step 4)**.** *`curl_curl_newtonVec`[coined]: for `IsTestOrbit` ω,* $\mathrm{curl}\,\mathrm{curl}(N \star \omega) = 4\pi\omega$, *using symmetry of second derivatives (`fderiv_fderiv_swap`),* $\mathrm{div}(N \star \omega) = N \star \mathrm{div}\,\omega = 0$, *and the step-3 theorem.*

Theorem 9. *`curlBiotSavartTest`[coined]: `CurlBiotSavartTest` holds, i.e.* $\mathrm{curl}\,\mathrm{BiotSavart}\,\omega = \omega$ *for every smooth, compactly supported, divergence-free ω. Consequently `helicity_casimir_test_closed`[coined]: helicity is a Casimir of the Arnold bracket at such ω, given the explicit argument $D\,S = u$.*

Scope: this is the classical identity on the smooth compactly supported class only; it says nothing about the dynamics. The Gâteaux derivative of helicity is treated in Section 8.

8 Helicity on the test class (`HelicityGateaux`)

Definition fix. `IsSmoothOrbit` previously used `ContDiff` $\mathbb{R}$ $\top$, which in current Mathlib means analytic; a compactly supported test orbit was then 0 and the earlier test-class statements were vacuous. It now uses C^∞ (the order $(\top : \mathbb{N}_\infty)$ coerced to `WithTop` $\mathbb{N}_\infty$); the class contains every $\mathrm{curl}\,\psi$ with $\psi \in C^\infty_c$ (`isTestOrbit_curlOrbit`[coined]).

Theorem 10. *`biotSavart_selfAdjoint`[coined]: for continuous compactly supported a, b, $\int \langle Ba, b\rangle = \int \langle Bb, a\rangle$ (Fubini with the majorant $|x-y|^{-2}|a(y)||b(x)|$, kernel symmetry, triple product).*

Theorem 11. *`helicity_gateaux`[coined]: with $\mathcal{H}(\omega) = \int \langle B\omega, \omega\rangle$ (`helicity`[coined]), $2B\omega$ is a Gâteaux representative of $\mathcal{H}$ on the test class.*

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Theorem 12. *curl_eq_of_isGateauxRepOn[coined]: two C^1 representatives of the same functional on the test class have equal curl. Hence helicity_casimir_of_rep[coined]: if $D\,S$ is any C^1 representative of $\mathcal{H}$, then $\operatorname{curl} D\,S = 2\omega$ and S is a Casimir of the Arnold bracket ArnoldBracketOf; no hypothesis on the value of $D\,S$.*

Theorem 13. *streamFunctionRepModGradOn_of_exists[coined]: StreamFunctionRepModGradOn follows from one C^2 kinetic-energy representative ψ with $\operatorname{curl}\psi = u$ and the Poincaré lemma CurlFreeIsGradient[coined]; the curl half of uniqueness is kineticEnergy_reps_curl_eq[coined].*

9 Poincaré lemma and the kinetic-energy representative (PoincareR3, KineticEnergyRep)

Theorem 14. *curlFreeIsGradient[coined]: every C^2 curl-free field on $\mathbb{R}^3$ is the gradient of a C^2 function.*

Theorem 15. *decays_newtonPotential[coined]: if g is continuous, compactly supported and $\int g = 0$, then $|N\star g(x)| \le C(1+|x|)^{-2}$. Derivatives of C^1_c functions and the coordinates of a divergence-free test field have mean zero (IsTestOrbit.integral_coord_eq_zero[coined]).*

Theorem 16. *integral_pairing_curl_symm[coined]: $\int\langle a, \operatorname{curl} b\rangle = \int\langle \operatorname{curl} a, b\rangle$ for fields with decaying coordinates and first derivatives. Hence, with $\psi = (4\pi)^{-1}N\star\omega$ (keStream[coined]), $\int\langle B\omega, B\varepsilon\rangle = \int\langle\psi,\varepsilon\rangle$, so ψ is a C^∞ Gâteaux representative of the kinetic energy with $\operatorname{curl}\psi = u$ (keStream_isGateauxRep[coined]).*

Theorem 17. *streamFunctionRepModGradOn[coined]: StreamFunctionRepModGradOn holds unconditionally on the test class.*

Smoothness audit. `instSmooth` in `Instance.lean` stated analyticity where C^∞ was meant (still true, now stated as C^∞). `contDiffAt_newtonRadial` states analyticity of $1/|x|$ off 0; that is true and proved, so it is left as is.

10 Maximizer data from a vorticity field (MaxPointDanskin, EntropyCasimirArn

Theorem 18. *danskin_upper_dini[coined]: if $f, \partial_t f$ are jointly continuous on $\mathbb{R}\times K$, K compact, then for some maximizer $x_\star$ of $f(t,\cdot)$ on K and every $\varepsilon > 0$, $\sup_K f(t+h) \le f(t,x_\star) + h(\partial_t f(t,x_\star) + \varepsilon)$ for small $h > 0$; i.e. $D^+\sup_K f(t) \le \partial_t f(t,x_\star)$. vorticity_danskin_upper_dini[coined] applies it to $|\omega|^2$.*

Theorem 19. *sup_mul_deriv_eq_inner[coined]: if $Y = \sup|\omega|$ is differentiable at t, then $YY' = \omega\cdot\partial_t\omega$ at every global maximizer. At such a point $\omega\cdot(v\cdot\nabla)\omega = 0$ (inner_convective_eq_zero_of_max[coined]) and $\omega\cdot\Delta\omega \le 0$ for C^2 fields (inner_laplacian_nonpos_of_max[coined]).*

Theorem 20. *maxPointRealization_of_field[coined]: for a solution of ModelVorticityPDE with $\nu \ge 0$, $\omega(t) \in C^2_c$, C^1 in time, and $Y = \sup|\omega| > 0$ differentiable, MaxPointRealization holds with EnergyIdentity, ViscousAtMax and CrossControlled discharged. The remaining hypotheses are FieldStretchingBound[coined] and FieldTetherFeedback[coined]. Running the ceiling argument from the one-sided Dini bound alone is open.*

Theorem 21. *arnoldCasimirField_helicity[coined]: $2u$ satisfies ArnoldCasimirField[coined] ($\forall a$, ArnoldBracketOf $\omega\, a\,(2u) = 0$) on the test class. This is the clean form of EntropyCasimir for the concrete bracket (entropyCasimir_of_arnoldCasimirField[coined]). The choice-based form EntropyCasimirChoice[coined] is kept and marked not provable as stated.*

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11 The ceiling from the Dini bound (`DiniCeiling`)

Theorem 22. *`ceiling_of_upper_dini`[coined]: if M is continuous on $[0,\infty)$, $M(t+h) \le M(t) + h(R(t)+\varepsilon)$ for small $h > 0$, and $R(t) < 0$ whenever $M(t) > B$, then $M(0) \le B$ implies $M \le B$ on $[0,\infty)$.*

Theorem 23. *`field_uniform_ceiling`[coined]: take a solution of `ModelVorticityPDE` with $\nu \ge 0$, $\kappa > 0$, C^2 in space, C^1 in time, and support in a fixed compact set. Under `FieldStretchingBound` and `FieldTetherFeedback`, $\sup|\omega(t)| \le \max(\sup|\omega(0)|, C/\kappa, 0)$ for all $t \ge 0$. No differentiability of $\sup|\omega|$ is assumed; its continuity is proved (`sup_continuous_of_uniform_support`[coined]).*

Theorem 24. *`fieldTetherFeedback_cubicTether`[coined]: for $T = -\kappa|\omega|^2\omega$ (`cubicTether`[coined]), `FieldTetherFeedback` holds. Hence `field_uniform_ceiling_cubicTether`[coined] needs only `FieldStretchingBo` For a general tether it does not follow.*

12 Status

- **Proved unconditionally:** `split_exists_unique`, `canonical_metric_bracket` (+ sign, symm/degen), energy/entropy structure, `uniform_ceiling`, `maxpoint_rate_logLoss` (as an implication from its listed Props), the obstructions above, the Newton potential identity (steps 1–3), `curlBiotSavartTest` (steps 4–5), Biot–Savart self-adjointness, the Poincaré lemma `curlFreeIsGradient`, `streamFunctionRepModGradOn` on the test class, $D\mathcal{H} = 2u$ and the helicity Casimir of the concrete Arnold bracket on the test class (`helicity_casimir_of_rep`), the scalar instance `instance_global_regularity`, the Danskin bound `danskin_upper_dini` and the maximum-principle lemmas.
- **Conditional on named Props:** `model_uniform_ceiling` (`TetheredMaxRate`); `model_global_regularity` (`TetheredMaxRate`, `ModelContinuation`); `model_global_regularity_of_realization` (`MaxPointRealiza` whose fields include `StretchingCZ` — false as an L^∞ NS bound); velocity-tether energy/entropy (`KineticEnergyCurlIdentity`, `EntropyCasimir`); helicity Casimir on the test class given $D\,S = u$ (`helicity_casimir_test_closed`; no longer conditional on `CurlBiotSavartTest`).
- **Open:** `EntropyCasimir` for the choice-based `FunctionalDerivative` bracket (not provable as stated); deriving the $|\omega|^2$ weight, κ and C3 from C1–C3; for the field-level realization (`maxPointRealization_of_field`): `FieldStretchingBound`, `FieldTetherFeedback` for tethers other than `cubicTether`, and existence and regularity (uniform compact support) of actual solutions. Differentiability of $\sup|\omega|$ is no longer needed (`field_uniform_ceiling`); and, for unmodified NS, the (H5) majorant, i.e. the Clay core. Neither the Clay problem nor NS regularity is claimed.
- The type named `T3` is $\mathbb{R}^3$, not the torus.

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